

Project Requirements

Project Title:

Adjustable Motor Mount for a NEMA 17 Stepper Motor

Project Goal:

Design a simple and adjustable motor mount for a NEMA 17 stepper motor. The mount should hold the motor securely and allow linear adjustment of the motor position for alignment or belt tensioning.

Main Requirements:

Requirement ID	Requirement	Target / Value	Verification Method
REQ-01	The motor plate shall interface with the simplified NEMA 17-style CAD model.	The 40 x 40 mm motor body is only a simplified CAD representation, not a verified commercial NEMA 17 envelope. The mounting interface uses four M3 holes on a 31 x 31 mm pattern. Final fit with a specific real motor requires verification.	Check the CAD assembly and mounting-hole alignment. Before manufacture, compare the drawing against the selected motor's datasheet and perform a physical fit check.
REQ-02	The motor plate shall retain the current compact geometry.	70 x 70 x 5 mm nominal.	Measure the CAD model or inspect the released drawing.
REQ-03	The base plate shall retain the current mounting envelope.	100 x 90 x 6 mm nominal.	Measure the CAD model or inspect the released drawing.
REQ-04	Each adjustment slot shall match the final CAD geometry.	6 mm width, 18 mm between end-arc centers, and 24 mm overall length.	Measure the slots in the CAD model or released drawing.
REQ-05	The motor position shall be linearly adjustable along one axis.	At least 18 mm of usable linear adjustment.	Measure the motor displacement between the two valid extreme positions in CAD or on a prototype.
REQ-06	Components shall remain clear throughout the full adjustment range.	No interference between the motor, plates, shaft, slots, or fasteners across the complete 18 mm travel.	Run a SolidWorks interference check at both travel limits and inspect intermediate positions.
REQ-07	The adjustment joint shall retain its selected position during representative bench operation.	No visible slip under a 30 N lateral static load. The 30 N load is a provisional portfolio design assumption, not a verified operating or safety rating.	Apply a 30 N lateral load to the mounted motor and compare its position before and after the test.
REQ-08	Structural stresses shall remain below the selected material's yield strength under the assumed load.	Calculated factor of safety ≥ 2 under the provisional 30 N lateral static load. The factor-of-safety target is a provisional portfolio design assumption and must be reassessed when actual operating loads are known.	Perform documented calculations using stated material properties, geometry, boundary conditions, and load direction.
REQ-09	The motor and adjustable joint shall use standard fasteners.	Four M3 motor fasteners and standard M5 adjustment	Inspect the CAD assembly, drawings, and BOM; verify

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		fasteners consistent with the current CAD components.	clearance and thread engagement.
REQ-10	The mount shall provide attachment points for connection to an external base.	At least four accessible base-mounting locations sized for the selected standard fasteners.	Inspect the base-plate drawing and perform a CAD or physical fit check with the intended support structure.
REQ-11	The custom plates shall be practical to manufacture using basic workshop processes.	Flat-plate construction using conventional cutting, drilling, and slot-machining operations; specialized processes must be documented if required.	Conduct a manufacturing review of the final part drawings.
REQ-12	Installation and adjustment shall be possible using common hand tools.	All required fasteners accessible with standard hex keys or wrenches; motor position adjustable without removing the motor from its plate.	Conduct an assembly and adjustment demonstration.
REQ-13	The released design shall have a minimum manufacturing and assembly drawing package.	Required drawing-package content: dimensioned drawings for the motor plate and base plate, plus an assembly drawing showing units, materials, general tolerances, fasteners, and adjustment direction.	Review the completed drawing package against a release checklist.

Selected Motor:

Simplified NEMA 17-style CAD representation

Initial Motor Data:

- Motor body size: 40 x 40 mm simplified CAD representation; not a verified commercial NEMA 17 envelope
- Mounting hole pattern: 31 x 31 mm
- Mounting holes: M3
- Shaft diameter: approximately 5 mm

Initial Design Dimensions:

- Motor Plate: 70 x 70 x 5 mm
- Base Plate: 100 x 90 x 6 mm
- Adjustment Slots: 6 mm wide, 18 mm between end-arc centers, and 24 mm overall length

- Standard fasteners will be used where possible

Planned Deliverables:

- SolidWorks part models
- Full assembly
- Engineering drawings
- Exploded view
- Bill of Materials (BOM)
- Basic GD&T
- Basic design calculations
- Final project documentation